QuantAnomalies

Quantitative Anomalies as Market Opportunities

Progress Presentation • Statistical & AI Dashboard for Equity and Forex Markets

1 Title of the Study

Quantitative Anomalies as Market Opportunities — A Statistical & AI Dashboard for Equity and Forex Markets.

> In plain terms: textbook finance says markets are "efficient" — prices already reflect everything, risk never changes, and nothing is predictable. Reality is messier, and those cracks in the theory are recurring patterns called **anomalies**. Our project hunts for those anomalies and reframes them as **opportunities**: places where the odds may tilt in a careful observer's favour.

2 Objectives of the Study

Each objective targets one well-known market anomaly and turns it into a usable opportunity signal.

1. **Macro-driver opportunity.** Measure *which* macroeconomic forces (interest rates, inflation, the fear index, oil, gold, the dollar) move an asset's returns — and crucially with *what time delay*. If a driver acts weeks later, that lag is a window to anticipate moves.
2. **Risk-regime opportunity.** Model how risk itself rises and falls over time (volatility clustering) and detect that crashes happen far more often than a bell curve predicts (fat tails). Spotting a "stormy" regime early is a chance to cut risk — or to recognise unusually cheap calm.
3. **Mean-reversion opportunity.** Find pairs of currencies whose prices are tethered over the long run (cointegration), then flag when they drift unusually far apart — history says the gap tends to snap back. This is the basis of market-neutral "statistical arbitrage".
4. **Opportunity synthesis (the recommendation engine).** Combine all of the above into one automated scanner that ranks what is unusual or actionable *right now*, and uses a **local AI analyst** to explain each finding in plain English.
5. **Accessibility & robustness.** Make it work for *any* ticker the user types, on a reliable data pipeline, presented through a clear, interactive dashboard.

3 Motivation for Selecting the Topic

- **The anomalies are real and well-studied, but rarely made tangible.** Lagged macro effects, volatility clustering, fat tails, and cointegration all appear in finance literature; we wanted to make them *visible and actionable* rather than abstract equations.
- **Reframing risk as opportunity.** The same statistics used to warn about risk can also point to opportunity — we wanted a tool that does both.
- **Bridging statistics and modern AI.** We combine classic econometrics (regression, GARCH, coin-

tegration) with a **local large language model** that turns numbers into a readable analyst note — private, free to run, and on our own GPU.

- **Learning value.** The project spans real statistical modelling, live financial-data engineering, full-stack web development, and on-device AI.

4 Data & APIs Used

In place of a questionnaire and a survey sample: our "instrument" is a set of financial-data APIs, and our "sample" is the live market history they return.

A. External data-source APIs (collecting raw market data)

- **Yahoo Finance** (via the `yfinance` library) — daily price history (2015–2025) for any equity, index, currency pair, future, or crypto. Examples: `^GSPC` (S&P 500), `^VIX` (volatility index), `CL=F` (oil), `GC=F` (gold), `^TNX` (10-year Treasury yield), `DX-Y.NYB` (US dollar index), and 45 forex pairs.
- **FRED — U.S. Federal Reserve Economic Data** — macroeconomic series: CPI (inflation), Fed Funds Rate, and Unemployment. A free mirror, **DBnomics**, is a built-in backup so a single outage never stops the study.

B. Our own analysis API (FastAPI backend)

Turns raw data into results:

- `/api/macro-regression/*` — regression, Granger-causality, correlation, time series
- `/api/garch/*` — volatility model, clustering tests, return distribution
- `/api/pairs/*` — cointegration tests, best-pair spread & signals, correlation
- `/api/recommendations` — the ranked opportunity scan
- `/api/llm/info` — status of the local AI analyst

C. Local AI API

An on-device `llama.cpp` server (OpenAI-compatible) running the **Qwen3-4B** model on our **NVIDIA RTX 4050 GPU**, used to explain detected opportunities in plain language. No data leaves the machine.

5 Pilot Study Conducted to Date

We have a **working end-to-end prototype**, validated on real data.

Data pipeline (validated). Live downloads work for all sources; a corrupted early cache (the volatility and oil series had been mixed up) was found and fixed. Every series now sits in its correct real-world range — e.g. VIX 9–83, oil -37 to 124, 10-year yield 0.5–4.9%.

Three analysis pillars — working, with sample findings on the S&P 500:

- *Macro:* macro factors explain about **64%** of monthly return variation ($R^2 \approx 0.64$); the model surfaces which factors are significant and at what lag.
- *Volatility:* GARCH "persistence" ≈ **0.9955** (shocks fade very slowly) and excess kurtosis ≈ **16** (strong fat tails) — risk is *not* constant and crashes are under-estimated by the normal curve.

- *Pairs:* among major forex pairs, **USDCHF/USDJPY** tested cointegrated ($p \approx 0.008$) with a mean-reversion half-life around 64 days.

Recommendation engine (v1 working). Four detectors (volatility regime, tail move, trend, forex mean-reversion) rank opportunities by severity and assign a confidence score. The local AI produces a grounded plain-English note in ≈4 seconds on the GPU (≈28 tokens/sec), citing only the real detected numbers.

Dashboard (working). Interactive web app: search any ticker, pick forex pairs, switch time ranges, and read explained results across five tabs (Overview, Opportunities, Macro, GARCH, Pair Trading).

> **In short:** the pilot confirms the data, the statistics, the AI-explanation layer, and the user interface all work together on live markets. Next steps (standardised comparisons, an options / Black–Scholes module, and a back-test of the signals) are tracked in the project roadmap.

6 Modules Built So Far

Backend (Python / FastAPI)

- `data_loader.py` — data acquisition (Yahoo Finance, FRED, DBnomics), caching, dataset builders
- `analysis/macro_regression.py` — Pillar 1: OLS lag regression, Granger causality, correlation
- `analysis/garch.py` — Pillar 2: GARCH(1,1), volatility clustering, return distribution
- `analysis/pairs.py` — Pillar 3: cointegration, spread / z-score, trading signals
- `analysis/recommender.py` — the opportunity engine: detectors, ranking, confidence
- `llm_client.py` — provider-agnostic local-LLM client (GPU server + CPU fallback)
- `main.py` — FastAPI endpoints tying it all together

Frontend (React + Vite)

- Five tab panels — Overview, Opportunities, Macro Regression, GARCH Volatility, Pair Trading
- Shared components — SVG icon set, info-tooltips, time-range filter, ticker search, forex selector, status states, animated background
- Support — chart theme, time-range utilities, data-fetch hook, API client

7 Future Plans — Expanding the Opportunity Engine

Each future module is another *lens* for spotting opportunity. The engine's principle stays the same: **statistics detect, the AI explains**.

1. Options & Black–Scholes

Price European call / put options and their Greeks, and back out the market's **implied volatility**. **Opportunity:** compare implied volatility with our GARCH forecast — when the market pays far more (or less) for risk than the model expects, options are "rich" or "cheap".

2. Valuation — "is the stock bloated / overpriced?"

Pull fundamental ratios (P/E, P/B, P/S, PEG, EV/EBITDA) and compare them with the stock's own history and its peers, alongside a technical "stretch" check (how far price sits above its 200-day average, and RSI). **Opportunity:** flag names "priced for perfection" (avoid / potential short) versus those "on sale" (potential value buys).

3. More opportunity lenses

- **Momentum & relative strength** — rank a universe by risk-adjusted 12–1 month momentum to spot rotation leaders and laggards.
- **Oversold bounce** — distance from the 52-week high plus RSI to find mean-reversion candidates.
- **Volatility squeeze** — detect compression that often precedes a breakout.
- **Correlation-regime shift** — assets that usually move together decoupling (a diversification or pairs opportunity).
- **Seasonality / calendar effects** — turn-of-month and "sell in May" style documented anomalies.
- **Alpha vs beta** — separate market-driven moves from stock-specific mispricing.
- **Unusual volume** — accumulation / distribution versus the average.
- **Risk budgeting** — turn the detected regime into a suggested position size.

8 Decision Engine: Rules-based vs LLM-based

We support **both**, and today combine them: *rules detect, the LLM explains*. A toggle lets the user choose the mode; pure rules is the safe default.

	Rules-based	LLM-based
Determinism	Fully deterministic, reproducible	Varies with settings
Explainability	Transparent thresholds	Natural-language reasoning
Numbers	Exact by construction	Must be guarded against errors
Flexibility	Rigid; new rules need code	Synthesises novel combinations
Speed & cost	Instant, free	≈4 s on GPU, compute cost
Works offline	Yes	Needs the local model
Best at	Detection & signals	Explanation & synthesis

> Our design uses each for what it is good at — rules produce the trustworthy numbers, the LLM produces the readable story. A future **LLM decision-maker** mode would let the model also weigh and rank signals (not just explain), kept honest by rule-based guardrails and number validation.

Appendix — Each anomaly in one line (layman)

- **Lagged macro effect:** *News ripples through markets over weeks, not instantly — know the lever and the delay, and you can position ahead.*
- **Volatility clustering:** *Calm days cluster, and so do wild days — storms arrive in groups, giving you warning to manage risk.*
- **Fat tails:** *Extreme crashes happen much more often than the textbook bell curve says — plan for them.*
- **Cointegration / pairs:** *Two currencies on one leash: when they drift far apart, they usually snap back — bet on the gap closing, up market or down.*